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Pavlovian bias is associated with symptom severity but not diagnostic status in individuals with both anxious and non-anxious depression

Carter M. Goldman[1], Navid Hakimi[1], Marishka M. Mehta[1,2], Maria Ironside[1,3], Martin P. Paulus[1,3] and Ryan Smith[1,3]✉

Human decision-making favors approach behavior to gain reward and avoidance behavior to prevent punishment. While adaptive in some cases, this *Pavlovian bias* can also interfere with instrumental motivation, leading to suboptimal decisions. As maladaptive avoidance is a known maintenance factor in anxiety and depression, this could be especially relevant to individuals with these disorders. To assess whether Pavlovian bias is altered in this population, we examined 106 healthy comparisons (HCs), 88 individuals with depression (Dep), and 184 with comorbid anxiety and depression (AnxDep) using an Orthogonalized Go/No-Go Task. Participants' choices and reaction times were modeled using a reinforcement learning and drift diffusion model (RL-DDM). As expected, results showed that accuracy was highest when Pavlovian and instrumental biases aligned and lowest when they conflicted. Linear models revealed no group differences in accuracy, reaction times, or any of the computational parameters. However, Pavlovian bias was positively associated with depression severity across individuals with both anxious and non-anxious depression. Anxiety sensitivity was also positively associated with Pavlovian bias in the AnxDep group specifically. Consistent with this, both depression severity and anxiety sensitivity in this group were also negatively associated with accuracy on the task when approach actions were required to avoid punishment. These results suggest that Pavlovian bias may contribute to symptom severity in both unique and overlapping ways within anxious vs. non-anxious depression. This may also specifically reflect suppression of approach behaviors when they would have adaptive value – potentially amplifying the avoidance behaviors known to maintain these disorders.

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INTRODUCTION

Human decision-making is known to favor approach behavior to achieve reward and avoidance behavior to prevent punishment [1–6]. This so-called *Pavlovian bias* is sometimes considered innate and appears separable from *instrumental* motivation, where one might instead learn that approach actions are most likely to avoid punishment or that avoidance is most likely to generate reward. Given that altered approach and avoidance behaviors are a hallmark of depression and anxiety disorders [7–10], prior work has begun to test the influence of Pavlovian bias on decision-making in these clinical groups and suggested potential relationships [5, 11, 12]. However, the cognitive mechanisms that govern Pavlovian bias and potential differences in individuals with anxiety and depression remain unclear. A better understanding of these mechanisms could potentially guide the development of more targeted interventions.

Computational models are useful tools for understanding the mechanisms that underlie decision-making and how they may be affected in individuals with affective disorders. *Reinforcement learning* (RL) is one common framework that can be used to describe the way previous observations inform the subjective value and choice of possible actions [13]. *Drift diffusion models* (DDMs) represent another prominent class of models that have been used to explain both choices and reaction times based on evidence accumulation toward a boundary that entails selection of one action vs. another [14]. These models have each previously been applied to paradigms that orthogonally manipulate instrumental learning and innate approach/avoidance drives to investigate how Pavlovian biases may interfere with instrumental control of behavior [5, 6, 15–18].

One established task used to assess the competition between Pavlovian and instrumental control is the Orthogonalized Go/No-Go Task [6]. This task measures the degree to which individuals can learn to select rewarding actions when they either do or do not conflict with Pavlovian biases. As expected, this paradigm has repeatedly shown that participants have more difficulty learning to choose avoidant actions to gain reward or approach actions to avoid punishment, as evident by lower accuracy and/or slower reaction times when instrumental learning and Pavlovian bias conflict [1, 5, 6, 16]. Both descriptive measures and computational modeling have also yielded specific insights regarding affective disorders in prior work. For example, one study found that participants with comorbid mood and anxiety disorders showed a greater tendency to inhibit behavior in the face of negative

[1]Laureate Institute for Brain Research, Tulsa, OK, USA. [2]College of Engineering and Natural Sciences, University of Tulsa, Tulsa, OK, USA. [3]Oxley College of Health and Natural Sciences, University of Tulsa, Tulsa, OK, USA. ✉email: rsmith@laureateinstitute.org

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outcomes than healthy participants [5]. Interestingly, they also found that the induction of state anxiety magnified this difference. While this finding corroborates the inhibitory component of anxiety disorders (e.g., patients with social anxiety disorder remaining withdrawn and quiet in social settings), it does not explain the active avoidance component (e.g., taking actions to avoid social settings altogether [19]). More in-line with this expectation, another study found that conditioned aversive stimuli had a greater effect on participants with major depressive disorder than healthy individuals, both increasing the tendency to actively avoid punishment and decreasing the tendency to approach rewards [11]. Overall, this suggests that depression and anxiety may relate to a heightened Pavlovian bias.

However, these findings have not been consistently replicated across studies [15, 16, 20, 21]. For example, one study using a Pavlovian-to-Instrumental Transfer Task found that, in healthy individuals, appetitive Pavlovian stimuli elicited greater active approach behavior, while aversive Pavlovian stimuli elicited greater active withdrawal responses [20]. In contrast, this action-specificity was not present in participants with major depressive disorder, suggesting reduced Pavlovian influence. Those with the greatest Pavlovian bias at baseline (i.e., most similar to healthy individuals) were also found to respond best to subsequent treatment. In two other studies, there was instead no relationship observed between appetitive or aversive Pavlovian bias (based on both choice behavior and reaction times) and affective symptoms [16, 21]. A further study found no difference in Pavlovian bias between individuals with major depressive disorder and matched controls. Thus, additional work is needed to resolve these inconsistent findings and confirm whether there is altered Pavlovian control in patients with affective disorders.

The relative contribution of anxiety and depression in this context also remains to be established. This is particularly relevant given recent work reporting qualitative behavioral differences in individuals with anxious vs. non-anxious depression. For instance, those with anxious depression have been shown to exhibit greater reactivity to both external threats and positive stimuli than individuals with depression alone [22, 23]. Diminished sensitivity to reward also appears more reflective of depressive than anxious symptomatology (for a review, see [24]). Despite strong evidence for these differences, their underlying computational mechanisms remain poorly understood.

To address these issues, in the present study we tested whether participants with comorbid anxiety and depression (AnxDep) might show differences in Pavlovian bias compared to those with depression alone (Dep), while also testing this bias in healthy comparisons (HCs) as a normative baseline. Models combining reinforcement learning and drift diffusion (RL-DDM) were fitted to participants' choices and reaction times on the Orthogonalized Go/No-Go Task, allowing measurement of individual differences in Pavlovian bias as well as a range of other computational mechanisms that could explain altered patterns of behavior. We compared groups based on fitted parameter estimates from the winning model. We then examined how descriptive and computational indices of behavior might account for continuous measures of depression and anxiety severity within the clinical groups.

As anxiety appears to be more associated with threat avoidance in the literature reviewed above, we hypothesized that, compared to the Dep group and HCs, the AnxDep group would show more frequent no-go responses (and slower go responses) in conditions where go responses were necessary to avoid punishment, consistent with an exaggerated Pavlovian bias. We expected Pavlovian influence on choice and reaction times would instead be similar between AnxDep and Dep in conditions where no-go responses were required to gain reward, as depression is present in both groups. We further tested whether this influence might be reduced in each group relative to HCs (e.g., similar to Huys et al. [20]), indicative of diminished approach motivation in depression. Within the clinical groups, we similarly hypothesized that aversive Pavlovian influences would be positively associated with anxiety severity, while appetitive Pavlovian influences might be negatively associated with depression severity.

METHODS

Recruitment

Participants were recruited from the community in and around Tulsa, Oklahoma, using social media posts, radio advertising, billboards, school messaging platforms, word-of-mouth, and other similar methods. Those interested were screened over the phone and in-person according to the Laureate Institute for Brain Research (LIBR) Screening Protocol approved by the Western Institutional Review Board (WIRB IRB # 20200161). Participants were required to be fluent in English and eligible for an MRI scan as part of a larger funded study. Participants were excluded for having a positive alcohol/drug test, neurocognitive disorder, traumatic brain injury, non-correctable vision, hearing problems, pregnancy, or one of the following DSM-5 disorders: schizophrenia spectrum and other psychotic disorders, bipolar disorder, or obsessive-compulsive disorder, as assessed by a clinical interview (Mini International Neuropsychiatric Interview 7 [MINI] [25]). In accordance with prior work utilizing this task [26], 6 individuals with major depressive disorder (Dep), 16 individuals with major depressive disorder and a comorbid anxiety disorder (AnxDep), and 7 healthy comparisons (HCs) were removed from the dataset due to indicators of poor task engagement (i.e., making the same choice for >90% of trials). The final dataset included 106 HCs (age: M=31.3, SD = 12.1), 88 Dep (age: M=33.8, SD = 12.1), and 184 AnxDep (age: M=31.6, SD = 11.1). Comorbid anxiety disorders in the AnxDep group included: generalized anxiety disorder (N = 131), social anxiety disorder (N = 93), panic disorder (N = 30), and agoraphobia (N = 10). Note that these categorizations are not mutually exclusive, as some participants had more than one anxiety disorder.

Protocol

Over multiple study visits at LIBR, participants completed a clinical interview and various medical, psychological, and demographic questionnaires (4.5 h; see Supplementary Methods for the psychometric properties of each questionnaire), an in-person interview (MINI) and computerized tasks (2.5 h), blood collection (15 min), and neuroimaging (4 h). Participants completed all sessions within a two-week period. Data were collected between 2018 and 2023. Data from blood collection, neuroimaging, and computerized tasks other than the Orthogonalized Go/No-Go Task are beyond the scope of the present work and will not be discussed further.

Orthogonalized Go/No-Go task

The Orthogonalized Go/No-Go Task (Fig. 1) is a probabilistic stop-signal task whereby participants are shown one of four stimuli (i.e., four shapes)

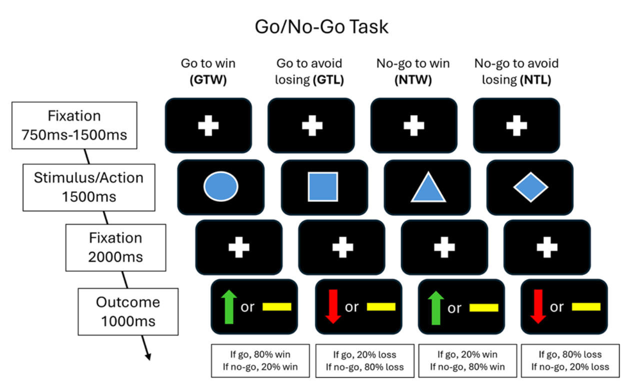

Fig. 1 Visual depiction of the Orthogonalized Go/No-Go Task. After a fixation lasting between 750 ms and 1500 ms, participants observed one of four shapes. Participants could emit a "go" response by tapping the spacebar within 1500 ms of stimulus onset or could emit a "no-go" response by allowing the time to elapse. After a 2000ms fixation, participants were shown a green upward-facing arrow (win), yellow horizontal bar (neutral), or red downward-facing arrow (loss) as a probabilistic result of their choice.

before deciding to go (i.e., press the spacebar) or no-go (i.e., not press anything) within a fixed time duration from trial onset [6]. Each stimulus corresponds to a different task condition determining the optimal action and outcome: 'go to win' (GTW), 'go to avoid losing' (GTL), 'no-go to win' (NTW), and 'no-go to avoid losing' (NTL). The GTW and NTW conditions yield only positive and neutral outcomes (GTW yields positive outcomes for go choices with 80% probability; NTW yields positive outcomes for no-go choices with 80% probability). The GTL and NTL conditions instead yield only negative and neutral outcomes (GTL yields negative outcomes for go choices with 20% probability; NTL yields negative outcomes for no-go choices with 20% probability). After a decision, participants observe a positive, neutral, or negative outcome (indicated by an upward-facing green arrow, horizontal yellow bar, or downward-facing red arrow, respectively) as a probabilistic result of their choice. Participants were told that they would receive a bonus between $5 and $55 depending on their performance – where positive outcomes added $1, negative outcomes subtracted $1, and neutral outcomes had no effect on the total bonus payment. They were also instructed that the outcomes associated with each stimulus were probabilistic and that outcome probabilities would remain stable throughout the task. The mapping from stimuli to task conditions was randomized for each participant. Participants completed 160 trials counterbalanced across each stimulus type, with a short break allowed every 40 trials.

Model

We performed Bayesian model comparison to assess eight RL models previously fit to choice data from this task (full modeling details are provided in Supplementary Methods, following the approach taken by Malamud, et al. [16]). In brief, these models calculated an action value for each possible action (i.e., go or no-go) given the observed stimulus. The simplest model only included a *go bias* parameter, which estimated a stable bias favoring go choices across the task for each participant. Further models added a parameter for Pavlovian bias, which increased the probability of go responses on reward trials and no-go responses on punishment trials. A *learning rate* determined the relative influence of new observations and prior experience when updating the expected instrumental value of go vs. no-go choices, while *outcome sensitivity* modulated the impact of each observed outcome. The overall action values, which integrated go bias, Pavlovian bias, and instrumental value, were then passed into a *softmax* function to calculate the probability of a particular action. This function also included a *decision noise* parameter to control choice stochasticity.

To assess our ability to differentiate data generated by each of the competing RL models, we conducted a model identifiability analysis that included simulating datasets under each model and then fitting those datasets to each model (see Supplementary Methods for details). Model comparison then confirmed whether the winning model in each case matched the model that was used to generate the data. This allowed us to verify that the empirical data was more likely generated by the winning model than alternative models tested, provided the winning model was not incorrectly identified as the best explanation for data generated by another model.

Building on prior work [16], we then fit participants' data to an additional set of models that combined the trial-by-trial action values learned within the winning RL model with a DDM model predicting both choices and reaction times (adapted from Millner, et al. [17]). The DDM models a stochastic evidence accumulation process during deliberation, where a choice is emitted when the evidence level crosses one of two decision boundaries (i.e., go and no-go). For each time step of the stochastic accumulation process, the level of evidence updates according to a Gaussian distribution parameterized by $N(v, \sigma)$. Here, v is referred to as the *drift rate* for the evidence, and σ is the drift variance, though the drift variance was fixed to 1 to isolate the effect of drift rate. The starting position of the evidence accumulation process (referred to as the *starting bias*) could be closer to either boundary, entailing a greater likelihood of reaching that boundary and shorter reaction times associated with that choice. The distance between boundaries (referred to as the *decision threshold*) can also differ by participant, where a greater decision threshold entails higher reaction times and a lower probability of hitting a boundary in the allotted time.

We tested nine combined RL-DDM models in which the trial-by-trial action values of the winning RL model were mapped onto various components of the DDM (see Supplementary Methods for complete model specifications). For these models, different subsets of the instrumental value, Pavlovian value, and go bias terms determined the drift rate and starting bias, while the decision threshold was fitted as a free parameter (i.e., as it does not entail a directional influence on choice consistent with the three RL terms). This round of model fitting also re-fit Pavlovian bias and the other RL model parameters to allow reaction time data to improve estimates of their underlying values. For go choices, the Wiener First Passage Time (WFPT) function calculated the probability density of a choice and reaction time combination as a function of the DDM parameters [27]. For no-go choices, we used the complement of the probability of the diffusion process hitting the go boundary (similar to the approach taken in previous work [17]). Posterior parameter estimates were hierarchically estimated using an established Parametric Empirical Bayes method combined with a variational Laplace fitting algorithm [28]. The protected exceedance probability for each model was estimated using Bayesian model selection to determine the winning model [29]. For the winning model, we assessed parameter recoverability by simulating data using plausible parameter estimates and then re-fitting the simulated data. The recoverability for each parameter was then measured as the correlation between the parameters used to simulate the data and the parameters estimated from fitting this simulated data (see Supplementary Methods for additional details regarding the parameter recoverability analysis). We also assessed our ability to identify data produced by each of the RL-DDM models by repeating the model identifiability analysis described above.

Measures

As our primary measures of depression and anxiety severity, respectively, participants completed the *Patient Health Questionnaire-9* (PHQ) [30] and *Overall Anxiety Severity and Impairment Scale* (OASIS) [31]. We also collected secondary measures for exploratory purposes. First, to assess the degree of concern elicited by specific anxiety symptoms, participants completed the *Anxiety Sensitivity Index-3* (ASI) [32], which included subscales for physical (ASI-Physical), cognitive (ASI-Cognitive), and social (ASI-Social) anxiety sensitivity. The *Ruminative Response Scale* (RRS) [33] was used to gauge brooding (RRS-Brooding), reflective thinking (RRS-Reflection), and depressive symptoms (RRS-Depression), which each constitute subscales of the measure. Levels of anhedonia and worry were measured by the *Snaith-Hamilton Pleasure Scale* (SHAPS) [34] and *Penn State Worry Questionnaire* (PSWQ) [35], respectively. The List Sorting Working Memory Test from the NIH Toolbox Cognition Battery was also used to assess working memory (note: the working memory scores we used here were based on *t*-scores adjusted for age and sex) [36].

Statistical analyses

Between-subject statistical analyses were carried out in R (version 4.2.1) with R Studio. Prior to running analyses, outliers were removed using an iterative Grubb's method (threshold: $p < 0.05$) using the *grubbs.test* function from the *outliers* package [37]. We removed outlier values for fitted parameter estimates, accuracy within each task condition (GTW, GTL, NTW, NTL), and mean reaction time within each condition separately to ensure that extreme values of those variables did not mask general patterns in the data. After removing outliers, linear regression models (LMs) and linear mixed-effect models (LMEs) were run using the *lm* function of the *stats* package and the *lmer* function of the *lme4* package, respectively [38, 39]. For the LMEs, participant ID was included as a random effect (i.e., allowing random intercepts). To further interpret null results, analogous Bayesian LMs and mixed effects models were run using the *lmBF* and *generalTestBF* functions of the *BayesFactor* package [40]. This allowed us to assess if the absence of a given predictor from a model better explained the data than the inclusion of that predictor. In other words, this tested whether evidence strongly favored a null model (as opposed to simply not being able to reject a null model). The Bayes Factor (BF) of a given predictor was calculated as the ratio of the likelihood of the model including that predictor to the likelihood of the model excluding that predictor.

Age and sex were included as covariates in all models to ensure they did not explain observed effects. When a significant effect of interest was observed, we subsequently tested models including working memory capacity as an additional covariate to evaluate whether it might account for the same variance as the predictor. We utilized this approach to make full use of the dataset, as 54 participants did not have available working memory scores. Thus, models including working memory required removal of these 54 participants, which led to a meaningful reduction in statistical power. Our main focus in these confirmatory analyses was therefore on comparison of changes in effect size when working memory was included.

All categorical variables were sum-coded (i.e., coefficients were estimated with respect to the value between levels of other categorical predictors). To aid in interpreting significant effects, estimated marginal means (EMMs) and estimated marginal trends (EMTs) were calculated using the *emmeans* and *emtrends* functions of the *emmeans* package [41], which allowed evaluation of post-hoc contrasts. All statistical tests were two-sided.

Descriptive analyses. The main descriptive metric of interest was task accuracy, which was separately calculated for each condition as the percentage of correct choices made. To evaluate if accuracy differed between diagnostic groups or task conditions, LMEs were run predicting accuracy based on condition, group, and their interaction. As a convergent test for group differences in task performance, LMs were also run that separately predicted accuracy within each task condition based on group. Analogous LMEs and LMs were run with reaction time as the outcome variable to determine if reaction times differed by group or task condition.

Relationships between descriptive indices of task performance and affective symptom severity (across all clinical participants, excluding HCs) were then assessed using separate LMs predicting accuracy in the two task conditions relevant to Pavlovian bias (GTL and NTW) based on PHQ and OASIS scores, respectively. A main effect of group was included to account for observed differences between Dep and AnxDep in these severity measures. An interaction between group and severity (PHQ or OASIS) was also included to evaluate whether the relationship between severity and task performance might differ between groups. After completing our primary hypothesis-driven analyses, we then carried out analogous tests on exploratory measures of the specific symptoms mentioned above: anxiety sensitivity (ASI), anhedonia (SHAPS), rumination (RRS), and worry (PSWQ). As these secondary analyses did not stem from a priori hypotheses, we utilized a Bonferroni adjustment for eight comparisons (i.e., two accuracy scores by four symptom scales). When results were significant for the total score on any of these measures, we also repeated analysis with each of its associated subscales. As this was done for interpretive purposes only (to better understand what drove the relationship with total scores), no correction for multiple comparisons was made. Otherwise, we applied a Bonferroni adjustment for the number of subscales tested. For the interested reader, we indicate when results were significant at both corrected and uncorrected levels.

Computational analyses. Between-subjects analyses were conducted using parameter estimates from the winning RL-DDM model. We first sought to investigate whether model fit differed between groups. Namely, an ANOVA was performed using the *aov* function of the *stats* package [42] that included group as a predictor of our fit metric (variational free energy), which considers both accuracy and complexity when evaluating the best model fit. As complementary tests, we also conducted the same test on the percentage of participants' choices that the model correctly predicted (i.e., to which it assigned the highest probability), and the average probability that the model assigned to each participant choice.

We next investigated how Pavlovian bias related to task performance. This allowed us to confirm whether higher values could be considered maladaptive in the context of the task. Specifically, an LME was run predicting task accuracy based on condition and Pavlovian bias. An interaction between these variables was also included to determine if certain values were more relevant to one condition than another. For a more complete understanding of the underlying determinants of task performance, similar analyses were also run for each of the other model parameters.

We also tested if groups showed altered decision-making during the task by comparing their levels of Pavlovian bias. This was done by running an LM predicting Pavlovian bias based on group. We next evaluated whether Pavlovian bias was associated with depression and anxiety severity across participants in both clinical groups together (i.e., excluding HCs). Specifically, separate LMs were run predicting Pavlovian bias values based on either PHQ or OASIS scores. These models included a main effect of group and an interaction between group and the relevant severity measure to assess if relationships might also differ between the Dep and AnxDep participants. We subsequently tested exploratory LMs with each of our secondary measures separately (ASI, SHAPS, RRS, and PSWQ) as predictors of Pavlovian bias. For these exploratory analyses, we utilized a Bonferroni adjustment for four comparisons (i.e., four scale measures). As above, when results for the total score of a given measure were significant, we repeated analyses for each associated subscale for interpretive purposes, and therefore did not apply further correction. When the effects for a total score were non-significant, we repeated analyses for each subscale while correcting for the number of subscales tested.

To determine if Pavlovian bias estimates might differ for participants using psychotropic medications, an LM was also run predicting Pavlovian bias values based on medication status (medicated/unmedicated) across all participants within the combined clinical groups. A similar LM was run to assess if accuracy in each condition related to medication status.

After all analyses of Pavlovian bias were complete, for exploratory purposes we re-ran each of these analyses for the other parameters within the winning RL-DDM model. These analyses were corrected for multiple comparisons based on the number of parameters and symptom measures (provided in Supplementary Results for the interested reader).

Because this task was administered as part of a larger funded study with other primary aims, a priori power analyses were not carried out for the specific analyses reported here. We therefore performed a sensitivity power analysis to evaluate the minimum effect size our sample would allow us to detect regarding the hypothesized group difference in Pavlovian bias (using the *pwr.anova.test* function of the *pwr* package [43]). Assuming a false positive rate of $p < 0.05$, this analysis indicated that our sample size of 378 would provide 80% power to detect a small-to-medium effect size of $\eta_p^2 = 0.025$.

RESULTS

As expected, compared to HCs, the Dep and AnxDep groups showed higher levels of depression, anxiety, anxiety sensitivity, anhedonia, rumination, and worry (see Table 1). Compared to Dep, AnxDep showed higher levels of anxiety, anxiety sensitivity (physical and social), rumination (brooding), and worry (see Supplementary Table S4 for pairwise comparisons). Based on established criteria [30, 44], there were 32 (36.4%) individuals in the Dep group and 62 (33.7%) individuals in the AnxDep group with PHQ scores between 5 and 9, indicating 'mild' depression; 29 (33.0%) individuals in the Dep group and 59 (32.1%) in the AnxDep group had scores between 10 and 14, indicating 'moderate' depression; 13 (14.8%) participants in the Dep group and 31 (16.8%) in the AnxDep group had scores between 15 and 19, signifying 'moderately severe' depression. Only 3 (3.4%) individuals in the Dep group and 14 (7.6%) in the AnxDep group had scores of at least 20, meeting the threshold for 'severe' depression.

Based again on previously established criteria [45], there were 39 (44.3%) individuals in the Dep group and 65 (35.3%) individuals in the AnxDep group with 'moderately severe' anxiety (OASIS score between 6 and 9); 15 (17.0%) individuals in the Dep group and 45 (24.5%) in the AnxDep group had 'marked' anxiety (OASIS score between 10 and 11); and 15 (17.0%) individuals in the Dep group and 53 (28.8%) in the AnxDep group had 'severe' anxiety (OASIS score greater than 12). There was not a significant difference in the proportion of individuals on psychotropic medication between the two clinical groups (see Table 1 for statistical results and Supplementary Table S5 for the proportion of individuals on different classes of psychotropic medication).

Accuracy and reaction times did not differ between groups

Initial outlier exclusion led to the removal of seven low accuracy values from the GTW condition (0–0.28), three high reaction time values from the GTW condition (1.14–1.21), two high reaction time values from the NTW condition (1.35, 1.39), one low reaction time from the GTL condition (0.25), and one high reaction time from the GTL condition (1.31).

The sufficient statistics for all behavioral measures are reported in Supplementary Table S6. These results confirmed that each group showed generally similar variances in each measure, including measures of Pavlovian bias, such as GTL Accuracy (HC: Mean=0.659, SD = 0.217; Dep: Mean=0.668, SD = 0.208; AnxDep: Mean=0.669, SD = 0.226), NTW Accuracy (HC: Mean=0.432, SD = 0.336; Dep: Mean=0.352, SD = 0.297; AnxDep: Mean=0.427, SD = 0.314), and the Pavlovian bias model parameter discussed further below (HC: Mean=0.428, SD = 0.671; Dep: Mean=0.529,

Table 1. Descriptive Statistics for Demographic and Clinical Measures.

Measure	*HCs (mean ± SD) N = 106*	*Dep (mean ± SD) N = 88*	*AnxDep (mean ± SD) N = 184*	*Statistical Test*	*p*	*Effect Size*
Sex	85 F, 19 M, 2 NA	64 F, 24 M	153 F, 31 M	$\chi^2(2) = 4.27$	0.118	$w = 0.11$
Age	31.31 (12.08)	33.83 (12.05)	31.60 (11.09)	$F(2, 375) = 1.39$	0.25	$\eta^2 < 0.01$
*Working Memory**	51.72 (9.54)	51.58 (10.31)	51.58 (9.00)	$F(2, 321) = 0.01$	0.993	$\eta^2 < 0.01$
*On Psychotropic Medication***	0 (0%)	27 (30.68%)	68 (36.96%)	$\chi^2(2) = 0.774$	0.379	$w = 0.053$
PHQ	1.40 (1.94)	10.05 (4.67)	11.07 (5.18)	$F(2, 375) = 174.98$	<0.001	$\eta^2 = 0.48$
OASIS	1.46 (2.48)	8.16 (3.49)	9.72 (3.28)	$F(2, 375) = 240.88$	<0.001	$\eta^2 = 0.56$
ASI-Social	4.44 (4.65)	11.10 (6.38)	13.45 (5.88)	$F(2, 375) = 85.24$	<0.001	$\eta^2 = 0.31$
ASI-Physical	1.57 (2.57)	5.28 (4.71)	7.02 (5.97)	$F(2, 375) = 41.00$	<0.001	$\eta^2 = 0.18$
ASI-Cognitive	0.93 (2.13)	7.12 (6.33)	7.68 (6.00)	$F(2, 375) = 58.76$	<0.001	$\eta^2 = 0.24$
RRS-Brooding	7.10 (2.22)	12.76 (3.29)	13.67 (3.43)	$F(2, 375) = 158.41$	<0.001	$\eta^2 = 0.46$
RRS-Reflection	7.17 (2.97)	11.92 (3.55)	11.67 (3.48)	$F(2, 375) = 71.00$	<0.001	$\eta^2 = 0.27$
RRS-Depression	16.31 (5.18)	31.77 (8.18)	33.10 (7.42)	$F(2, 375) = 205.80$	<0.001	$\eta^2 = 0.52$
SHAPS	0.67 (2.13)	2.24 (2.74)	2.35 (2.52)	$F(2, 375) = 16.98$	<0.001	$\eta^2 = 0.08$
PSWQ	35.89 (11.72)	57.08 (14.33)	65.05 (10.62)	$F(2, 375) = 204.40$	<0.001	$\eta^2 = 0.52$

For each measure, the mean and standard deviation (SD) are provided for each group. The right-most columns show the results of statistical tests for group differences, including the test statistic, p-value, and effect size.

PHQ patient health questionnaire, *OASIS* overall anxiety severity and impairment scale, *ASI-Social*, *ASI-Physical*, *ASI-Cognitive* social, physical, and cognitive subscales of the anxiety sensitivity index, *RRS-Brooding*, *RRS-Reflection*, *RRS-Depression* brooding, reflection, and depression subscales of the ruminative response scale, *SHAPS* Snaith-Hamilton pleasure scale, *PSWQ* penn state worry questionnaire.

*Note only 88 HCs, 76 Dep, and 160 AnxDep had data for Working Memory.

***On Psychotropic Medication* indicates whether the participant was currently taking one or more of the following types of medication: Selective Serotonin Reuptake Inhibitors, Serotonin-Norepinephrine Reuptake Inhibitors, Norepinephrine-Dopamine Reuptake Inhibitors, Anxiolytics, Mood Stabilizers, Stimulants, Tricyclics. Given that no participants in the HC group were currently taking psychotropic medication, only the clinical groups were considered in the statistical test for a difference in this measure.

SD = 0.661; AnxDep: Mean=0.453, SD = 0.670). Visual inspection of the residual distributions for these measures in all main analyses further confirmed that the assumptions of each statistical test were met.

Our first LME predicting task accuracy based on group, condition, and their interaction revealed a significant effect of condition ($F(3,1111.2) = 291.03$, $p < 0.001$, $\eta_p^2 = 0.44$; see Fig. 2), but no effect of group ($F(2,370.4) = 1.47$, $p = 0.231$, $\eta_p^2 < 0.01$) or interaction ($F(6,1111.0) = 1.33$, $p = 0.242$, $\eta_p^2 < 0.01$). The effect of condition reflected that accuracy was highest in the GTW condition (EMM = 0.90), not significantly different between the GTL (EMM = 0.68) and NTL conditions (EMM = 0.69), and lowest in the NTW condition (EMM = 0.42; see Supplementary Table S7 for post-hoc contrasts). Greater accuracy in the GTW than GTL condition is consistent with the idea that instrumental control was more effective when congruent with Pavlovian bias (i.e., as both favor approach behavior in GTW only). Greater accuracy in the NTL than NTW condition supports the same interpretation, as avoidance (no-go) responses are favored by both instrumental and Pavlovian control for NTL only. Additionally, accuracy negatively related to age ($b = -0.002$, $F(1,370.7) = 13.83$, $p < 0.001$, $\eta_p^2 = 0.04$) and was higher for males (EMM[male]=0.70, EMM[female]=0.65, $F(1,373.0) = 8.99$, $p = 0.003$, $\eta_p^2 = 0.02$). These effects remained present when working memory was included in the model and without outlier removal (see Supplementary Results). Subsequent Bayesian analyses confirmed that there was strong evidence for models without a main effect of group ($BF_{10} = 0.042$) and very strong evidence for models without an interaction between group and condition ($BF_{10} = 0.0086$) [46]. Further, for LMs that separately predicted accuracy values for each condition based on group, there was not a significant effect of group in any case ($ps > 0.134$).

An analogous LME predicting reaction times based on group, condition, and their interaction also showed a significant effect of condition ($F(3,1104.6) = 123.84$, $p < 0.001$, $\eta_p^2 = 0.25$; see Fig. 2), but no effect of group ($F(2,370.7) = 0.43$, $p = 0.653$, $\eta_p^2 < 0.01$) or interaction ($F(6,1104.3) = 0.81$, $p = 0.565$, $\eta_p^2 < 0.01$). The main effect of condition revealed that reaction times were slower in the GTL condition (EMM = 0.80) than in the GTW condition (EMM = 0.70), and faster for error responses in the NTW (EMM = 0.76) than NTL conditions (EMM = 0.82). Reaction times were higher in males than females (EMM[male]=0.79, EMM[female] =0.75, $F(1,372.6) = 5.56$, $p = 0.019$, $\eta_p^2 = 0.01$) and positively related to age ($b = 0.002$, $F(1,370.7) = 11.65$, $p < 0.001$, $\eta_p^2 = 0.03$). These effects remained significant when working memory was included as a covariate and without the removal of outliers (see Supplementary Results for the results of these models).

Pavlovian influences on task accuracy were associated with symptom severity within the clinical groups

The next set of LMs predicted accuracy for the GTL and NTW conditions separately within clinical participants only (i.e., excluding HCs), while including primary symptom severity measures (PHQ or OASIS), group, and their interaction as predictors. In the model predicting GTL accuracy, we observed a significant negative association with PHQ ($b = -0.007$, $F(1,266) = 9.08$, $p = 0.003$, $\eta_p^2 = 0.03$; see Fig. 3) and OASIS ($b = -0.008$, $F(1,266) = 5.65$, $p = 0.018$, $\eta_p^2 = 0.02$; see Fig. 3), though only the former remained significant when working memory was added to the model (PHQ: [$b = -0.005$, $F(1,229) = 5.66$, $p = 0.018$, $\eta_p^2 = 0.02$], OASIS: [$b = -0.005$, $F(1,229) = 3.26$, $p = 0.072$, $\eta_p^2 = 0.01$]). To disentangle effects of depression and anxiety, we then included both variables as predictors of GTL accuracy in the same LM. This revealed a main effect of PHQ ($b = -0.006$, $F(1,266) = 4.36$, $p = 0.038$, $\eta_p^2 = 0.02$), but no effect of OASIS ($b = -0.005$, $F(1,266) = 0.99$, $p = 0.321$, $\eta_p^2 < 0.01$). Given the strong relationship between anxiety and depression symptoms, we also confirmed that variance inflation factors for PHQ (1.37) and OASIS (1.42) were acceptable, minimizing potential concerns about multicollinearity. These results jointly suggest depression

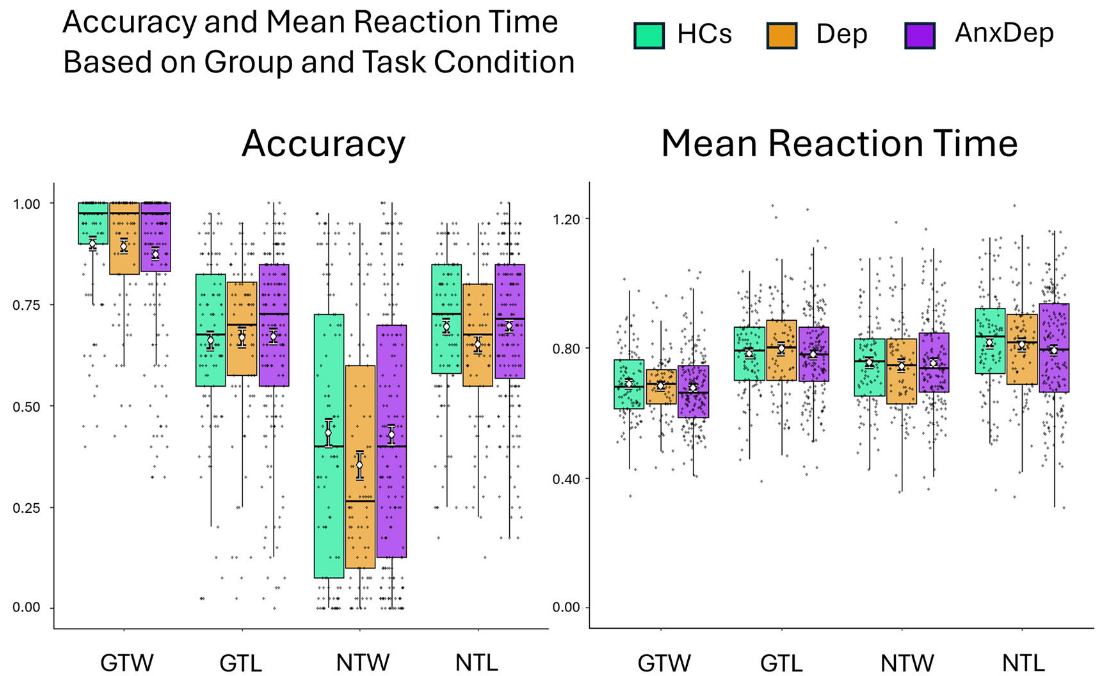

Fig. 2 Accuracy and mean reaction time for each group and task condition. Boxplots show the median and quartile values in addition to data points. The white circles and surrounding bars indicate the mean and standard error estimates. Accuracy was highest in the 'go to win' (GTW) condition, not significantly different between the 'go to avoid losing' (GTL) and 'no-go to avoid losing' (NTL) conditions, and lowest in the 'no-go to win' (NTW) condition. Reaction times were slower in the GTL condition than in the GTW condition, and faster for error responses in the NTW than NTL conditions (i.e., as reaction times cannot be calculated for no-go responses). This suggests less conflict between instrumental and Pavlovian control in the GTW than GTL condition, and more impulsively driven errors in the NTW than NTL condition – both consistent with the direction of Pavlovian bias.

severity contributed to lower performance in the GTL condition. In the model predicting NTW condition accuracy including OASIS scores, we also observed an effect of group (EMM[Dep]=0.36, EMM[AnxDep]=0.44, $F(1,266) = 4.18$, $p = 0.042$, $\eta_p^2 = 0.02$), suggesting that Dep participants may have struggled more to inhibit behavior in expectation of reward. However, given that we did not observe group differences in our initial group analyses above, we simply note this result here for the purpose of future hypothesis generation.

Subsequent exploratory analyses predicting task accuracy in the GTL and NTW conditions using secondary measures showed a number of significant effects. First, we observed a significant interaction between ASI and group ($F(1,266) = 7.75$, $p = 0.006$, $\eta_p^2 = 0.03$) and a main effect of ASI ($b = -0.002$, $F(1,266) = 8.68$, $p = 0.003$, $\eta_p^2 = 0.03$) predicting GTL accuracy (see Fig. 3); both effects survived correction for eight comparisons. When working memory was added to the model, only the interaction remained significant after correction (ASI x Group: [$F(1,229) = 7.80$, $p = 0.006$, $\eta_p^2 = 0.03$], ASI: [$b = -0.0002$, $F(1,229) = 6.30$, $p = 0.013$, $\eta_p^2 = 0.03$]). To better interpret these relationships, we then tested additional LMs including ASI-Social, ASI-Cognitive, and ASI-Physical. Here, we observed negative relationships between GTL accuracy and both ASI-Physical ($b = -0.003$, $F(1,266) = 7.05$, $p = 0.008$, $\eta_p^2 = 0.03$) and ASI-Cognitive ($b = -0.004$, $F(1,266) = 6.46$, $p = 0.012$, $\eta_p^2 = 0.02$). There was also a significant interaction between ASI-Social and group ($F(1,266) = 6.22$, $p = 0.013$, $\eta_p^2 = 0.02$), indicating that the relationship between social anxiety sensitivity and GTL accuracy was negative in the AnxDep group (EMT $= -0.008$), but positive in the Dep group (EMT $= 0.003$). A similar interaction was observed for ASI-Physical ($F(1,266) = 4.82$, $p = 0.029$, $\eta_p^2 = 0.02$) and was marginal for ASI-Cognitive ($F(1,266) = 3.67$, $p = 0.056$, $\eta_p^2 = 0.01$). These findings suggested that the interaction between anxiety sensitivity and group may not be specific to any particular domain of anxiety sensitivity.

Although no significant results were found in relation to RRS total scores, further exploratory analyses revealed that RRS-Brooding scores were negatively related to GTL accuracy ($b = -0.008$, $F(1,266) = 9.67$, $p = 0.002$, $\eta_p^2 = 0.04$) and moderated the effect of group ($F(1,266) = 6.78$, $p = 0.010$, $\eta_p^2 = 0.02$), suggesting a negative association in AnxDep but a slightly positive association in Dep. Both results survived correction for three comparisons. After working memory was added to the model, we observed marginal evidence for the main effect of RRS-Brooding and significant evidence for the moderation effect with group (RRS-Brooding: [$b = -0.005$, $F(1,229) = 5.34$, $p = 0.022$, $\eta_p^2 = 0.02$], RRS-Brooding x Group: [$b = -0.010$, $F(1,229) = 5.84$, $p = 0.016$, $\eta_p^2 = 0.02$]). Similar to the interaction observed for anxiety sensitivity, these results indicated that the relationship between RRS-Brooding and GTL accuracy was negative in AnxDep (EMT $= -0.019$), but positive in Dep (EMT $= 0.003$; see Fig. 3). Taken together, these results suggest that brooding related to lower performance on the task in conditions with Pavlovian effects, especially for individuals with anxious depression. In all other models, no effects were observed for group, symptoms, or their interaction. Medication status was also not a significant predictor of task accuracy in any condition ($ps > 0.223$).

All analogous tests predicting reaction times in the GTL and NTW conditions based on PHQ and OASIS scores revealed non-significant results (for details, see Supplementary Tables S9 and S10). Exploratory analyses with secondary symptom measures also did not show any significant effects (see Supplementary Tables S11 and S12).

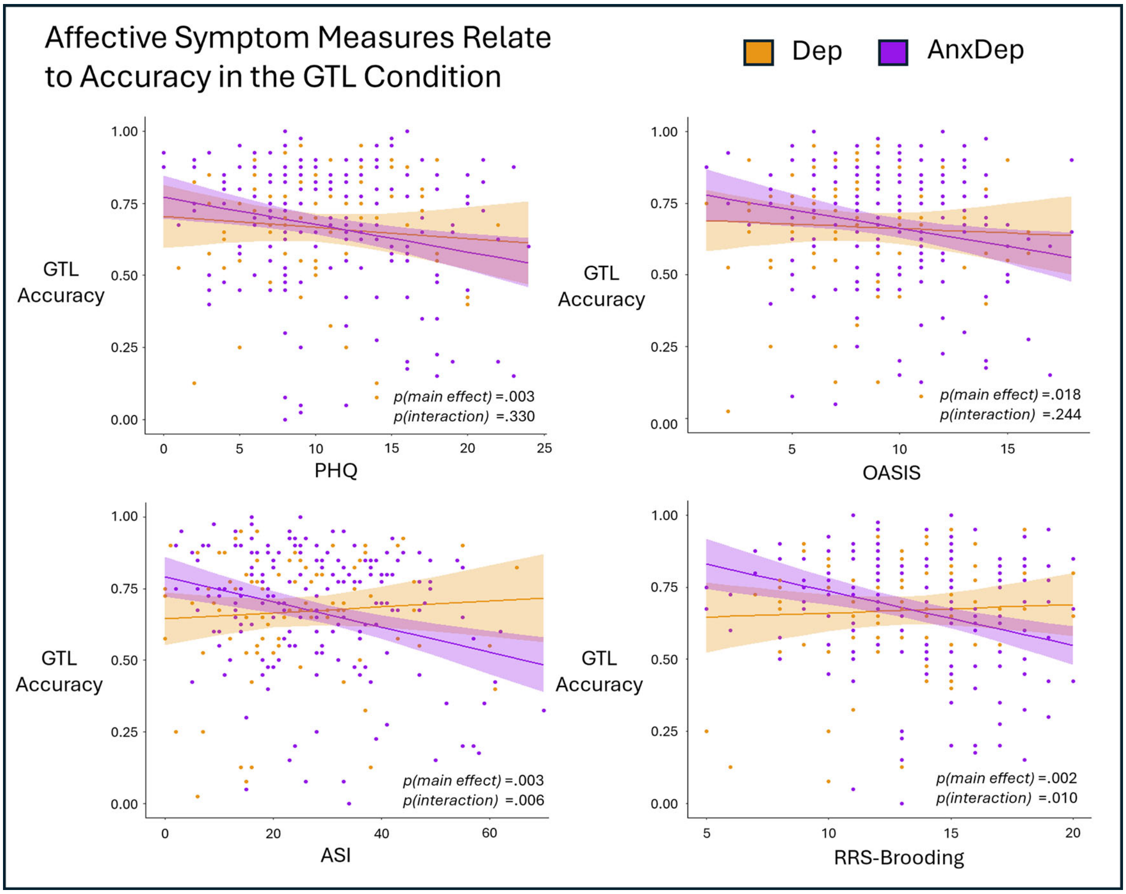

Fig. 3 Regressions predicting GTL accuracy based on affective symptom measures within the combined clinical groups. *Top-left*: Patient Health Questionnaire (PHQ) scores negatively related to accuracy in the 'go to avoid losing' (GTL) condition. *Top-right*: Overall Anxiety Severity and Impairment Scale (OASIS) scores also negatively related to GTL accuracy. *Bottom-left*: Anxiety Sensitivity Index (ASI) scores related to GTL accuracy in a group-specific manner. *Bottom-right*: the Brooding subscale of the Ruminative Response Scale (RRS) also related to GTL accuracy in a group-specific manner. Shaded areas indicate confidence intervals for each trend line. The terms *p(main effect)* and *p(interaction)* indicate the *p*-value associated with the predictor variable on the x-axis and the interaction between that variable and group, respectively. These models included age and sex as covariates.

Model comparison and validation

Across the space of RL models fitted to participant data, model comparison selected the model including one outcome sensitivity, learning rate, go bias, Pavlovian bias, and decision noise parameter (protected exceedance probability [*pxp*]=1). Identifiability analyses confirmed that this model was not mis-identified when data were generated by simpler models. However, it was identified as the best model for data generated by more complex models that further separated parameters between rewarding and punishing conditions (see Supplementary Table S13 for detailed results). Thus, while it is unlikely that simpler models generated our data, we cannot rule out that our participants' behavior was influenced by more complex valence-dependent mechanisms. Because the winning model remained the simplest plausible explanation of our data, we focus on this model going forward.

When subsequently testing the RL-DDM models (nine in total), the winning model indicated that instrumental value, Pavlovian value, and go bias jointly governed the drift rate only ($pxp = 1$). In other words, the summation of instrumental value, Pavlovian value, and go bias influenced the speed and direction of the evidence accumulation process, where higher values entailed a greater probability of making a "go" decision and faster reaction times. This model additionally included free DDM parameters for the starting bias and decision threshold. The model correctly predicted 79% of choices across all participants, indicating acceptable accuracy. We did not observe group differences in our main model fit metric (variational free energy; HCs = −72.90, Dep = −75.70, AnxDep = −73.31, $F(2,375) = 0.185$, $p = 0.831$, $\eta_p^2 < 0.01$); nor did we observe differences in model accuracy (HCs = 0.792, Dep = 0.787, AnxDep = 0.791, $F(2,375) = 0.055$, $p = 0.947$, $\eta_p^2 < 0.01$) or the probability that the model assigned to chosen actions (HCs = 0.703, Dep = 0.694, AnxDep = 0.704, $F(2,375) = 0.299$, $p = 0.742$, $\eta_p^2 < 0.01$). Identifiability analyses showed that data generated by alternative RL-DDM models were not misattributed to the winning model within model selection, confirming that the winning model was most likely to have generated the empirical data (see Supplementary Table S14 for detailed results). All model parameters were also recoverable (see Supplementary Figures S1 and S2 for full results of the parameter recoverability analysis).

Prior to analyzing the relationship between computational parameters and other behavioral as well as psychological measures, a small number of outliers were removed (see **Methods** for additional details). This included one high learning rate value (0.81), three high outcome sensitivity values (4.53–13.76), eight low starting bias values (0.28–0.44), and four high decision threshold values (4.65–6.42). We next tested LMEs predicting accuracy based on Pavlovian bias, as well as based on the other model parameters separately. Here, we found that lower Pavlovian bias predicted better task performance, as expected (full statistical details shown in Supplementary Table S15). Higher values of outcome sensitivity and decision threshold also predicted greater

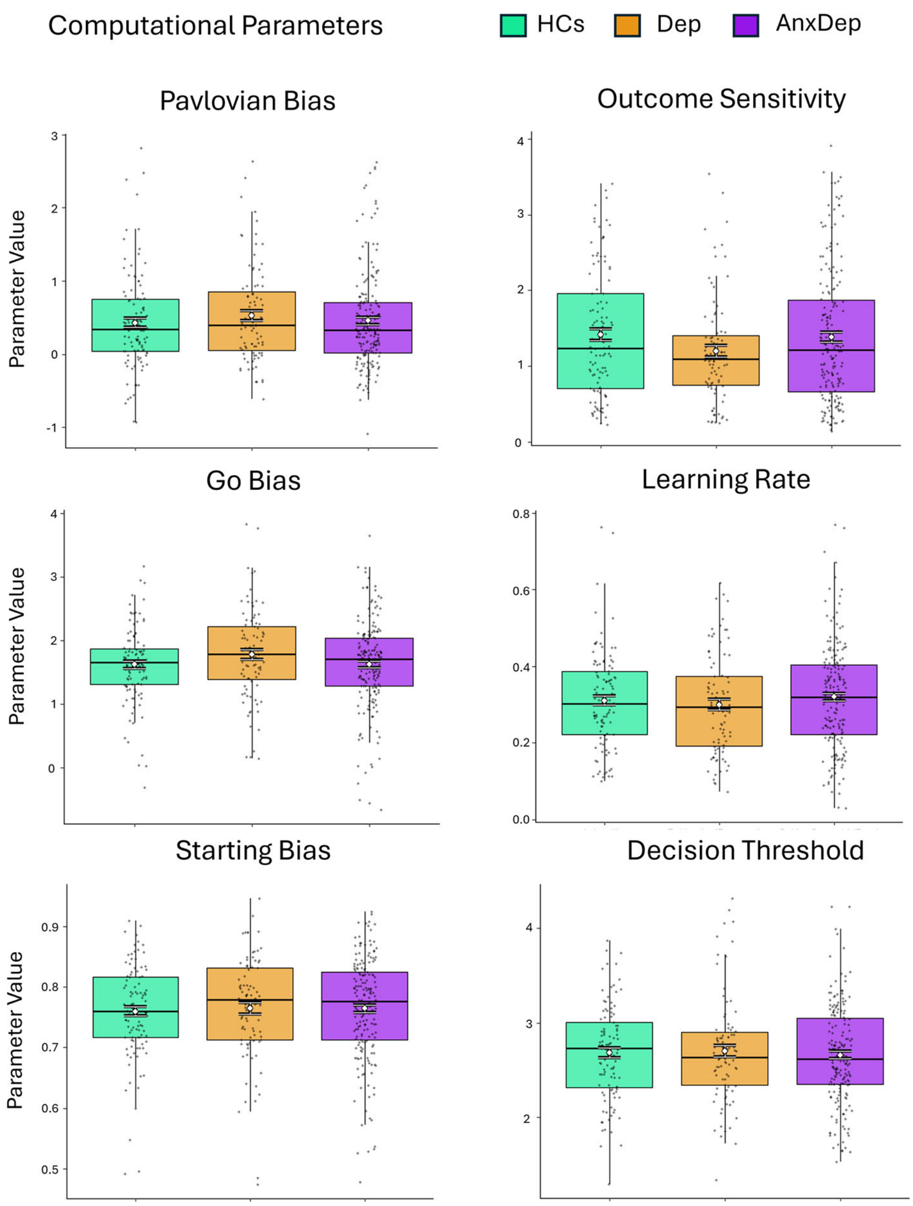

Fig. 4 Computational parameter values separated by group. These results supported the absence of a group difference for each computational parameter. Boxplots show the median and quartile values in addition to single data points. Overlaid white circles and error bars indicate means and standard errors.

performance. Conversely, lower values of starting bias predicted greater performance. There were also several significant interactions in which the relationship between accuracy and computational parameters differed between conditions. These patterns confirmed that each model parameter accounted for behavior in the expected manner.

Pavlovian bias does not differ between groups but is associated with symptom severity

Initial LMs did not reveal significant group differences for Pavlovian bias or any other model parameter (see Fig. 4). We also did not observe any group differences when working memory was included as a covariate (see Supplementary Table S16 for full statistical results). Further, analogous Bayesian models of Pavlovian bias indicated that the data were roughly 10 times more likely under a model without a group difference than a model including this difference ($BF_{10} = 0.082$, $F(2,316) = 0.60$, $p = 0.548$, $\eta_p^2 < 0.01$), which is typically taken to reflect a strong level of evidence [46]. When outliers were included, we again observed no group differences (full statistical results provided in Supplementary Results).

Subsequent LMs separately predicted Pavlovian bias based on measures of depression or anxiety severity, accounting for effects of group and their potential interaction. Here, we observed an effect of PHQ ($b = 0.014$, $F(1,266) = 4.82$, $p = 0.029$, $\eta_p^2 = 0.02$; see Fig. 5). This positive association between Pavlovian bias and depression was no longer significant when working memory was added to the model ($b = 0.008$, $F(1,229) = 1.80$, $p = 0.181$, $\eta_p^2 < 0.01$). However, working memory was not itself associated with Pavlovian bias ($b = -0.002$, $F(1,229) = 0.25$, $p = 0.618$, $\eta_p^2 < 0.01$). Interestingly, when follow-up models were run within each group separately, we observed an increased effect size for PHQ in AnxDep ($b = 0.022$, $F(1,180) = 5.55$, $p = 0.020$, $\eta_p^2 = 0.03$), but no effect for Dep ($b = 0.006$, $F(1,84) = 0.16$, $p = 0.691$, $\eta_p^2 < 0.01$). Given the moderately negative correlation between

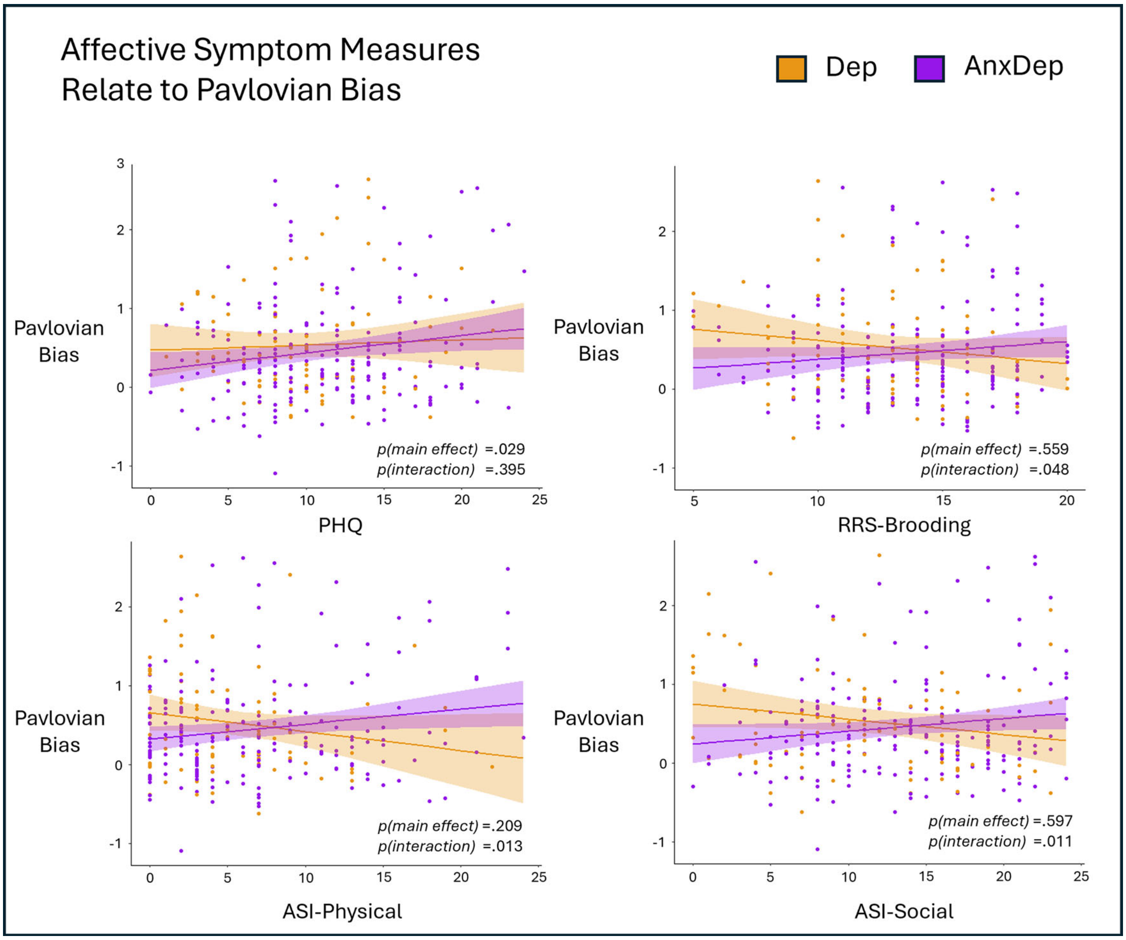

Fig. 5 Regressions predicting Pavlovian bias based on affective symptom measures within the combined clinical groups. Pavlovian bias was positively related to Patient Health Questionnaire (PHQ) scores (top-left). Groups differed in the association between Pavlovian bias and the Brooding subscale of the Ruminative Response Scale (RRS-Brooding; top-right), the physical anxiety subscale of the Anxiety Sensitivity Index (ASI-Physical; bottom-left), and the social anxiety subscale of the Anxiety Sensitivity Index (ASI-Social; bottom-right). Shaded areas indicate confidence intervals for each trend line. The terms *p(main effect)* and *p(interaction)* indicate the *p*-value associated with the predictor variable on the x-axis and the interaction between that variable and group, respectively. These models included age and sex as covariates. Note that the interaction effect of RRS-Brooding and group predicting Pavlovian bias did not survive correction for multiple comparisons.

Pavlovian bias and learning rate ($r = -0.419$, $t(375) = -8.95$, $p < 0.001$; see Figure S3 for correlations between all RL-DDM parameters), we next investigated whether the link between Pavlovian bias and depression level (PHQ) might be accounted for by differences in learning rate. Here, we found that when including both parameters as regressors in an LM, each had non-significant effects on PHQ (Pavlovian bias: $b = 0.584$, $F(1,265) = 1.32$, $p = 0.252$, $\eta_p^2 < 0.01$; learning rate: $b = -4.463$, $F(1,265) = 3.25$, $p = 0.072$, $\eta_p^2 = 0.01$), suggesting that these parameters may explain partially shared variance in depression severity.

In exploratory analyses predicting Pavlovian bias based on our secondary symptom measures, we also identified some noteworthy results. First, we observed an interaction between group and ASI ($F(1,266) = 6.14$, $p = 0.014$, $\eta_p^2 = 0.02$). Although this initial test did not survive correction for four comparisons, it did survive correction in the subsequent model controlling for differences in working memory ($F(1,229) = 8.52$, $p = 0.004$, $\eta_p^2 = 0.04$). To interpret this interaction, we tested separate regressions for each subscale. This revealed that both social and physical anxiety sensitivity were higher in those with greater Pavlovian bias in AnxDep (EMT[ASI-Social]=0.016, EMT[ASI-Physical]=0.019; see Fig. 5), while the opposing association was present in Dep (EMT[ASI-Social]=−0.019, EMT[ASI-Physical]=−0.024). Further exploratory analyses revealed a similar interaction with RRS-Brooding ($F(1,266) = 3.93$, $p = 0.048$, $\eta_p^2 = 0.01$; see Fig. 5), although this effect did not survive correction for multiple comparisons. This effect was also not significant when working memory was included in the model ($F(1,229) = 2.99$, $p = 0.085$, $\eta_p^2 = 0.01$). Medication status did not significantly predict Pavlovian bias ($F(2,268) = 0.25$, $p = 0.617$, $\eta_p^2 < 0.01$), nor did it relate to any other model parameter ($ps > 0.231$). In exploratory analyses testing symptom severity relationships with each of the other computational parameters, we observed no significant associations at corrected levels (see Supplementary Tables S17–S22).

DISCUSSION

In this study, we used computational modeling to evaluate whether Pavlovian bias differed in individuals with anxious vs. non-anxious depression (AnxDep vs. Dep), and whether Pavlovian bias might further predict symptom severity in one or both groups. Contrary to our hypothesized group effects, there was strong evidence favoring the absence of a difference in Pavlovian bias between AnxDep and Dep, and no differences between those groups and healthy comparisons (HCs). This suggested that mechanisms underlying Pavlovian bias may be robust and not directly influenced by the presence of these affective disorders, consistent with some prior results in the mixed literature on this task to date (such as absences of group differences between major depression and matched comparisons in prior work [15, 16, 21]). In contrast, we did observe significant continuous associations between Pavlovian bias and symptom severity across the Dep and AnxDep groups. In particular, we found that Pavlovian bias was greater in those with higher depressive symptom severity. This was in the opposite direction of our initial hypothesis, which assumed higher depression severity would relate to a reduced go-bias under reward-seeking conditions. However, it does suggest Pavlovian bias could be a clinically meaningful predictor or act as a vulnerability or maintenance mechanism for depression. On the other hand, we did observe stronger Pavlovian influence on GTL accuracy in those with higher depression and anxiety severity, and exploratory analyses also suggested a consistent positive relationship between Pavlovian bias and anxiety sensitivity that was specific to the AnxDep group. Thus, only results for anxiety were in expected directions, but unexpected relationships with depression nonetheless supported a type of reduced approach behavior. We discuss these results in more detail below.

Initial results indicated that participants' choices were best explained by an RL-DDM model that included one free parameter for Pavlovian bias, as opposed to separable biases in rewarding and punishing conditions. This suggests that individuals with Dep vs. AnxDep may not have differed in this bias when seeking reward vs. avoiding threat. Analyses of descriptive measures of task behavior were also consistent with this conclusion, as groups did not differ in GTL or NTW conditions. However, it is worth noting that model identifiability analyses showed that simulated data generated by models with separate free parameters for rewarding and punishing conditions were also best fit by the model with a single Pavlovian bias parameter. Thus, we cannot rule out that distinct mechanisms were present in controlling appetitive and aversive Pavlovian influences on behavior. Future work might further consider how each of these influences separately contribute to specific deficits in anxiety and depression. For example, perhaps difficulty with acting to avoid punishment could be more strongly associated with maladaptive avoidance, while difficulty inhibiting action to gain reward could be more associated with non-reflective or impulsive choice.

In contrast to the observed absence of group effects, Pavlovian bias was found to positively predict depressive symptom severity (PHQ) across the clinical samples. This was consistent with the pattern of associations found with GTL accuracy, more specifically suggesting difficulty initiating approach behaviors to avoid punishment. Here, levels of anxiety (OASIS) also related to lower GTL accuracy; however, when PHQ and OASIS scores were included in the same model, only PHQ scores retained a significant effect. These results are consistent with prior work finding reduced approach motivation in depression more generally [47–51]. However, our results more specifically suggest that greater depressive symptoms may be associated with an amplified bias to overwrite prior reinforcement of approach behaviors to avoid loss. While we initially expected to find a weaker Pavlovian bias in reward-seeking conditions, this result is consistent with forms of maladaptive avoidance in depression often targeted in transdiagnostic treatment approaches [52, 53].

Subsequent exploratory results using narrower dimensional symptom measures also found interesting results worth further investigation in future work. In particular, higher brooding-related rumination (RRS-Brooding), social anxiety sensitivity (ASI-Social), and physical anxiety sensitivity (ASI-Physical) were each related to greater Pavlovian bias in the anxious depression group, with an opposing relationship in those with non-anxious depression. Here, it is notable that the AnxDep group showed higher symptom scores than the Dep group for these measures (see Supplementary Results). Thus, it is possible that individuals with the most severe symptoms were more likely to follow innate approach-avoidance drives despite interference with task performance. Alternatively, it could be that the combination of anxiety and depression yields cognitive biases that prevent individuals from approaching goals under negatively valenced states. Additional work will be needed to confirm these findings and understand how they relate to real-world dysfunction. It should also be noted that not all results found for brooding survived correction for multiple comparisons and should be interpreted with caution.

It is important to consider some further limitations of the present study when interpreting the overall pattern of results described above. Firstly, we did not have a non-depressed anxious group. This prevented us from determining whether individuals with primarily anxious symptoms might show maladaptive Pavlovian biases. Future work will be needed to investigate if relationships between Pavlovian bias and affective symptoms are present in such individuals. It is also important to note that only three participants in the Dep group and fourteen in the AnxDep group met cutoffs for severe depression. We therefore cannot confirm the absence of group differences in Pavlovian bias in populations with higher depression severity. Another important consideration is that recruitment for this study was restricted to a single location within the United States. Future studies will be needed to determine if these findings generalize to other geographic regions and cultures. Given that different studies have made use of different tasks, another constraint of the present study was the use of a single task. It may be that studies simultaneously collecting data from multiple Pavlovian bias tasks in the same individuals will be necessary to further resolve discrepancies in this literature. For example, in one prior study discussed above [20], a Pavlovian-to-instrumental transfer task was used, with results instead suggesting less Pavlovian influence in a co-morbid depressed sample relative to HCs. Yet, Pavlovian bias in this study was instead operationalized as the influence of conditioned stimulus-outcome associations, which is notably different from its operationalization here in terms of the congruency between approach-avoidance and win-loss conditions. If future work were to show distinct vs. similar associations with Pavlovian bias in these two tasks within the same participants, this would lend important insights into how apparent inconsistencies in prior work would be best interpreted.

While the winning model performed well — correctly predicting 79% of participants' choices — it is also important to highlight that other prominent modeling approaches have received limited attention to date. In particular, Bayesian inference models represent another framework with additional explanatory resources that might offer novel insights regarding mechanisms underlying Pavlovian biases and could be explored in future studies (e.g., see [54, 55]). Another modeling-related limitation stems from the fact that Pavlovian bias and learning rate estimates were significantly correlated and appeared to explain overlapping variance in depression levels. Thus, while our hypotheses were specific to Pavlovian bias, this result limits our ability to determine the relative contribution of these parameters with high confidence. A further consideration along these lines stems from how working memory demands in the task were accounted for in the winning model, which could bear on interpretation of estimated learning rates. Here, this parameter is used to explain both the impact of new information and the loss of old information (e.g., forgetting due to working memory limitations). Yet, processes underlying gaining and losing information might also be distinct. While we used an established space of RL models for this task based on previous studies, future work might therefore benefit from incorporating alternative modeling approaches that separate learning and forgetting mechanisms. With that said, working memory scores were tested as covariates in analyses above, which we expect at least partially addressed potential confounds that could have arisen from individual differences in this ability.

In summary, we did not find a difference in Pavlovian bias between individuals with anxious depression, depression alone, or healthy comparisons. However, we did find evidence for a positive association between Pavlovian bias and depression severity across those with both anxious and non-anxious depression. Exploratory results further suggested that anxiety sensitivity was related to higher Pavlovian bias in those with anxious (but not non-anxious) depression, most prominently when the task required approach behavior to prevent negative outcomes (with a similar suggestive pattern in relation to rumination). Subsequent work will need to assess whether these results generalize to real-world settings. If so, Pavlovian bias might represent a novel target for future treatment studies.

DATA AVAILABILITY

All data used in the analyses reported in this manuscript are available at: https://github.com/cgoldman123/Hierarchical_Reinforcement_Learning_Drift_Diffusion_Model_Go_NoGo_Task.

CODE AVAILABILITY

All code used for analyses reported in this manuscript is available at: https://github.com/cgoldman123/Hierarchical_Reinforcement_Learning_Drift_Diffusion_Model_Go_NoGo_Task.

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AUTHOR CONTRIBUTIONS

C.M.G. and R.S. led the data analysis and manuscript preparation. C.M.G., N.H., M.M.M., M.I., M.P.P., and R.S. contributed to the review and editing of the manuscript. M.P.P. also led the larger funded study in which these data were collected.

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COMPETING INTERESTS

The authors have no competing interests to declare.

ETHICS AND CONSENT

The study protocol was approved by the Western Institutional Review Board (WIRB #20182352) and conducted in accordance with the Declaration of Helsinki. All subjects provided written informed consent prior to participation.

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Correspondence and requests for materials should be addressed to Ryan Smith.

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